EC536 Applied Econometrics

PS4 Q1

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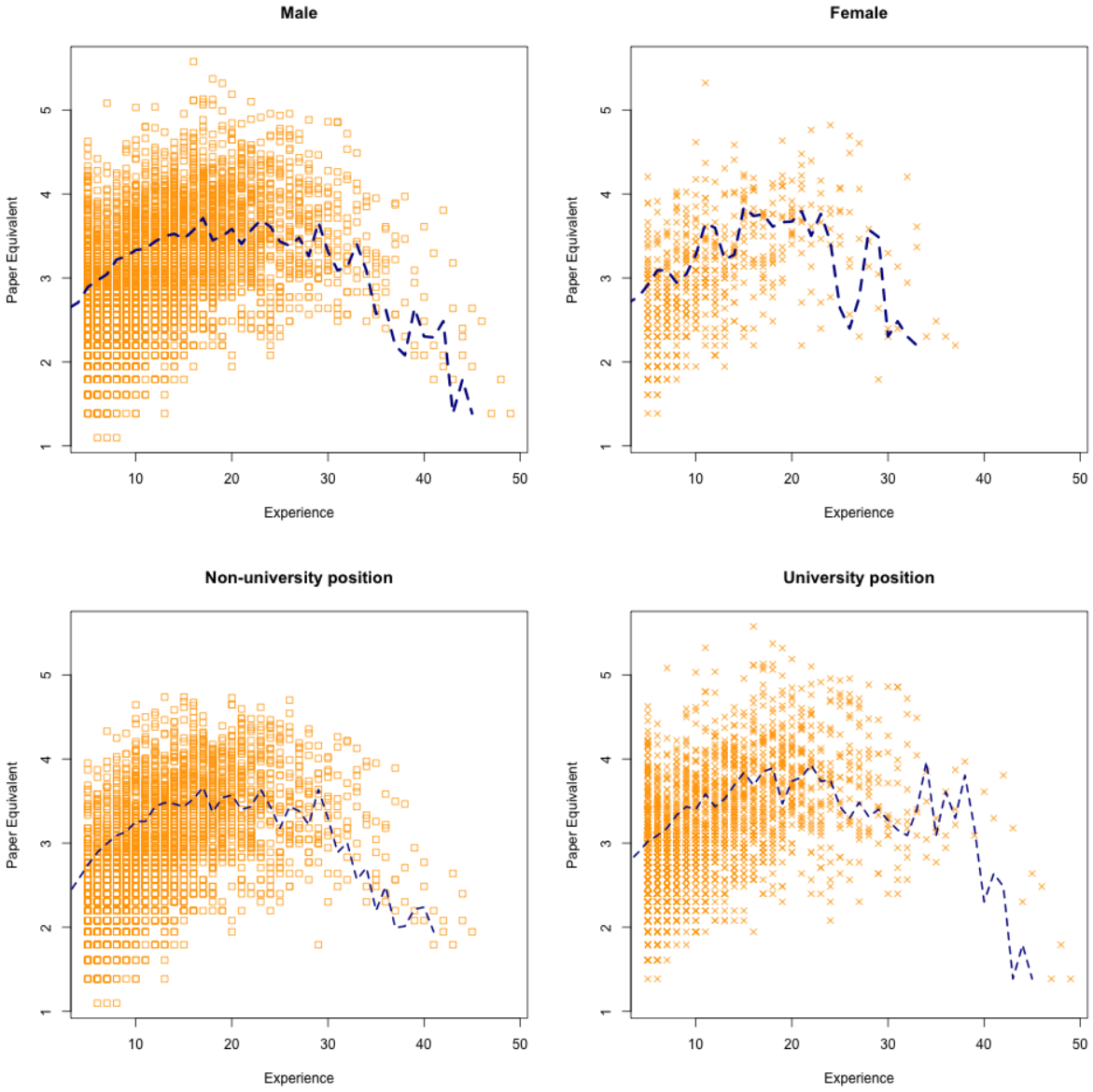
Q1 - Shape of productivity curve



Regression results

$$\log y_{it} = \beta_0 + \sum_{j=1}^{2} \rho_j \log y_{it-j} + \beta_1 e_{it} + \beta_2 e_{it}^2 + \beta_3 \frac{1}{e_{it} r_i} + \beta_4 F_i + \beta_5 D_i + \alpha_i + u_{it}$$

	(1) Fixed	(2) Between	(3) Random	(4) Hausman-Taylor
$\log y_{it-1}$	0.543^{***}	1.134^{***}	0.679^{***}	0.543^{***}
	(0.013)	(0.034)	(0.012)	(0.012)
$\log y_{it-2}$	-0.426^{***}	-0.393^{***}	-0.333^{***}	-0.426^{***}
	(0.012)	(0.033)	(0.012)	(0.012)
e_{it}	0.155^{***}	0.070^{***}	0.113^{***}	0.155^{***}
	(0.006)	(0.008)	(0.004)	(0.006)
e_{it}^2	-0.003^{***}	-0.002^{***}	-0.002^{***}	-0.003^{***}
	(0.0001)	(0.0003)	(0.0001)	(0.0001)
$\frac{1}{e_{it} r_i}$	2.183^{***}	0.045	1.187^{***}	2.174^{***}
	(0.447)	(0.186)	(0.172)	(0.439)
F_i		0.008	-0.023	-0.013
		(0.020)	(0.021)	(0.040)
D_i		0.031	0.004	0.010
		(0.028)	(0.015)	(0.031)

Conclusions

- Hausman statistic: reject null, so favour FE
  - Use HT model
  - e_{it}^2 term has negative coefficient
- Other specifications did not improve model much:
  - Including y_{it-3}, $\frac{1}{r_i}$
- Productivity curve has concave shape